

ENGINEERING PHYSICS

Anna University | Complete Exam-Ready Answer Book

PART B — 13 Mark Questions

Full Scoring Model Answers

Prepared strictly as per Latest Anna University Syllabus & Valuation Pattern

TOPICS COVERED

1. Torsional Oscillations & Rigidity Modulus
2. Air Wedge
3. Michelson Interferometer
4. Nd:YAG Laser
5. CO₂ Laser
6. Solid State, Gas & Semiconductor Laser Comparison
7. Optical Fibre Communication System
8. Ultrasound Imaging Systems
9. Time-Dependent Schrödinger Wave Equation
10. Particle in an Infinite Potential Well (Quantization of Energy)
11. Transmission Electron Microscope (TEM)

Q1. Torsional Oscillations & Rigidity Modulus using Torsional Pendulum [13 Marks]

■ 1. Definition

Torsional Oscillation is the to-and-fro angular (twisting) motion of a body suspended by a wire or rod. When the body is twisted and released, it oscillates due to the restoring torque produced in the wire. A Torsional Pendulum uses this phenomenon to determine the Rigidity Modulus (Modulus of Rigidity) of the material of the wire.

■ 2. Principle

When a cylindrical wire is twisted through an angle θ , a restoring torque is developed in the wire that is proportional to the angle of twist. The body suspended by the wire undergoes Simple Harmonic Motion (angular SHM). By measuring the time period of oscillation and knowing the dimensions of the wire and disc, we can calculate the rigidity modulus.

■ 3. Construction (Torsional Pendulum)

A Torsional Pendulum consists of the following parts:

- **Wire / Rod:** A thin metallic wire of known length (L) and radius (r) whose rigidity modulus is to be determined.
- **Suspended Disc:** A heavy circular disc of known mass (M) and radius (R) is fixed to the lower end of the wire.
- **Upper Clamp:** The upper end of the wire is fixed rigidly to a support.
- **Pointer and Scale:** A pointer attached to the disc and a circular scale to measure the angle of twist.

■ 4. Diagram — Torsional Pendulum

[illegible]

■ 5. Theory and Working

When the disc is twisted by a small angle θ and released, the wire exerts a restoring torque (τ) on the disc:

$$\tau = -\mathbf{C} \cdot \boldsymbol{\theta}$$

Where C is the torsional rigidity (restoring couple per unit angular twist) of the wire.

By Newton's second law for rotational motion:

$$I \cdot (d^2\theta/dt^2) = -C \cdot \theta$$

This is the equation of angular SHM. Comparing with standard SHM equation:

$$\omega^2 = C / I \rightarrow \text{Time Period } T = 2\pi \sqrt{I/C}$$

Where I = Moment of Inertia of the disc about the axis of the wire.

◆ Torsional Rigidity (C) of the Wire

From the theory of elasticity, the torsional rigidity of a cylindrical wire is:

$$C = (\pi \eta r^4) / (2L)$$

Where:

- η = Rigidity Modulus of the wire material (to be found)
- r = Radius of the wire
- L = Length of the wire

◆ Moment of Inertia of the Disc

$$I = M R^2 / 2$$

Where M = Mass of disc, R = Radius of disc

■ 6. Derivation of Rigidity Modulus

From the time period formula:

$$T = 2\pi \sqrt{I/C}$$

Squaring both sides:

$$T^2 = 4\pi^2 (I/C)$$

Therefore:

$$C = 4\pi^2 I / T^2$$

Substituting $C = (\pi \eta r^4)/(2L)$ and $I = MR^2/2$:

$$(\pi \eta r^4)/(2L) = 4\pi^2 \times (MR^2/2) / T^2$$

Solving for η (Rigidity Modulus):

$$\eta = (8\pi L M R^2) / (r^4 T^2) \leftarrow \text{FINAL FORMULA}$$

◆ Symbol Summary

Symbol	Meaning	Unit
η	Rigidity Modulus	N/m ²
L	Length of wire	m
M	Mass of disc	kg
R	Radius of disc	m
r	Radius of wire	m
T	Time Period of oscillation	s

■ 7. Procedure

- Measure the length (L) and radius (r) of the wire using a screw gauge and scale.
- Measure the mass (M) and radius (R) of the disc.
- Twist the disc slightly (small angle) and release it.
- Measure the time for 20 oscillations and find time period $T = \text{Total time} / 20$.
- Substitute all values in $\eta = 8\pi LMR^2 / rT^2$ to get rigidity modulus.

■ 8. Advantages

- Simple and accurate method.
- Only small angular displacements are needed.
- Can be used for wires of any metallic material.
- Non-destructive testing method.

■ 9. Disadvantages

- Damping due to air resistance can affect accuracy.
- Wire must be uniform throughout its length.
- Accurate measurement of wire radius is difficult (r term — small error gives large error).

■ 10. Applications

- Determination of rigidity modulus of metals like steel, copper, aluminium.
- Used in material testing in engineering industries.
- Design of shafts and springs in mechanical engineering.

■ 11. Important Exam Points

- ★ The final formula is: $\eta = 8\pi LMR^2 / rT^2$
- ★ Torsional oscillation is angular SHM — restoring torque is proportional to angle of twist.
- ★ Units of rigidity modulus η are N/m^2 or Pascal (Pa).
- ★ r appears in the formula — small error in r gives large error in η .

■ 12. Conclusion

The Torsional Pendulum is an elegant and accurate experiment to determine the rigidity modulus of a material. By measuring the time period of torsional oscillations and the physical dimensions of the wire and disc, we can compute η using the formula $\eta = 8\pi LMR^2 / rT^2$.

Q2. Air Wedge — Principle, Construction, Working and Applications [13 Marks]

■ 1. Definition

An Air Wedge is a thin air film of gradually increasing thickness formed between two glass plates, with one end in contact and the other end separated by a thin wire or paper. It is used to determine the diameter of a thin wire or the wavelength of light using the principle of interference of light.

■ 2. Principle

Air Wedge works on the principle of **Interference of Light** (specifically, interference in thin films).

When monochromatic light falls on the air wedge, it partially reflects from the top surface of the air film (Ray 1) and partially from the bottom surface (Ray 2). These two reflected rays travel slightly different paths (path difference) and interfere with each other.

- Where the path difference = $n\lambda \rightarrow$ **Constructive Interference** $\rightarrow$ Bright fringe
- Where the path difference = $(2n+1)\lambda/2 \rightarrow$ **Destructive Interference** $\rightarrow$ Dark fringe

This produces alternating bright and dark fringes (interference fringes).

■ 3. Construction

- Two optically flat glass plates are placed one on top of the other.
- One end of the plates is in perfect contact (or tied together).
- At the other end, a thin object (wire or paper) of diameter/thickness 'd' is placed.
- This creates a wedge-shaped air film between the plates with wedge angle θ .
- A monochromatic light source illuminates the system from above through a glass plate at 45° .

■ 4. Diagram — Air Wedge Setup

[illegible]

■ 5. Theory and Working

◆ Path Difference

For a ray reflecting from the air film of thickness 't' at a point:

$$\text{Path Difference} = 2t + \lambda/2$$

(The extra $\lambda/2$ is due to phase change of π upon reflection at the denser glass plate at the bottom.)

◆ Condition for Dark Fringes (Destructive Interference)

$$2t = n\lambda \quad (n = 0, 1, 2, 3, \dots)$$

◆ Condition for Bright Fringes (Constructive Interference)

$$2t = (2n + 1) \lambda/2 \quad (n = 0, 1, 2, 3, \dots)$$

◆ Fringe Width (β)

The wedge angle $\theta = d/L$ where d = diameter of wire, L = distance from contact edge to wire.

The fringe width (spacing between consecutive dark or bright fringes) is:

$$\beta = \lambda / (2\theta) = \lambda L / (2d)$$

◆ Finding Diameter of Wire

From the fringe width formula, the diameter of the wire is:

$$d = \lambda L / (2\beta) \leftarrow \text{IMPORTANT FORMULA}$$

Where:

- λ = Wavelength of monochromatic light used
- L = Distance from contact edge to the wire
- β = Fringe width measured under microscope
- d = Diameter of wire

■ 6. Nature of Fringes

- Fringes are **straight, parallel, and equally spaced**.
- At the contact edge ($t = 0$): Always a **dark fringe** (due to the $\lambda/2$ phase shift).
- Fringes are perpendicular to the length of the wedge.
- Fringes become closer as wedge angle increases.
- If white light is used, coloured fringes are obtained (central fringe is dark).

■ 7. Advantages

- Non-contact, non-destructive method of measuring small diameters.
- Very accurate — can measure diameter in micrometres.
- Simple and inexpensive setup.

■ 8. Disadvantages

- Requires monochromatic light source for clear fringes.
- Glass plates must be optically flat.
- Vibration can disturb the fringes.

■ 9. Applications

- Measurement of the diameter of thin wires or fibres.
- Testing the flatness of optical surfaces.
- Determination of wavelength of light.

- Measurement of thickness of thin films, papers, or sheets.
- Quality control in optics manufacturing.

■ 10. Important Exam Points

- ★ **Key formula:** $d = \lambda L / 2\beta$
- ★ **At contact edge:** DARK fringe (not bright) due to $\lambda/2$ phase change.
- ★ **Fringes are straight, parallel and equally spaced** — unlike Newton's rings which are circular.
- ★ **Air wedge is a special case of thin film interference.**

■ 11. Conclusion

The Air Wedge experiment beautifully demonstrates the wave nature of light through thin film interference. It is a simple, accurate, and non-destructive technique used to measure the diameter of thin wires and test the flatness of surfaces. The formula $d = \lambda L / 2\beta$ is the key result of this experiment.

Q3. Michelson Interferometer — Principle, Construction, Working, Nature of Fringes and Applications [13 Marks]

■ 1. Definition

The Michelson Interferometer is an optical instrument invented by Albert A. Michelson in 1881. It uses the principle of division of amplitude to produce two coherent beams from a single light source, creates an interference pattern, and is used for precise measurements of wavelength, wavelength difference, and refractive index.

■ 2. Principle

Michelson Interferometer works on the principle of **Division of Amplitude**. A single beam of light is split into two beams by a semi-silvered mirror (beam splitter). These two beams travel different paths, reflect from mirrors, and recombine to interfere. The interference depends on the path difference between the two beams.

■ 3. Construction

◆ Key Components:

- **Light Source (S):** Monochromatic or white light source.
- **Beam Splitter (G1):** A semi-silvered glass plate inclined at 45° — splits the incoming beam into two: reflected (Beam 1 $\rightarrow$ M1) and transmitted (Beam 2 $\rightarrow$ M2).
- **Compensating Plate (G2):** A glass plate identical to G1 but without silver coating, placed parallel to G1 in the path of Beam 2 to equalize the optical path lengths.
- **Mirror M1:** A fixed plane mirror perpendicular to Beam 1.
- **Mirror M2:** A movable plane mirror perpendicular to Beam 2 (can be moved using a fine screw with micrometer accuracy).
- **Observer / Telescope (E):** At the output to observe interference fringes.

■ 4. Diagram — Michelson Interferometer

[illegible]

Interference fringes seen at E

■ 5. Working

Step 1: Light from source S falls on beam splitter G1 (at 45°).

Step 2: G1 splits the beam into two — Beam 1 (reflected towards M1) and Beam 2 (transmitted towards M2).

Step 3: Both beams hit their respective mirrors (M1 and M2) and reflect back to G1.

Step 4: The two beams recombine at G1 and proceed to the observer E.

Step 5: Since the two beams originate from the same source (coherent), they interfere and produce fringes at E.

Step 6: When M2 is moved, the path difference changes → fringes shift. By counting fringe shifts, precise measurements are made.

■ 6. Path Difference and Fringe Conditions

Let 'd' be the difference in the distances of M1 and M2 from the beam splitter G1.

$$\text{Path Difference} = 2d$$

Conditions:

$$\text{Bright fringe: } 2d = n\lambda \quad (n = 0, 1, 2, \dots)$$

$$\text{Dark fringe: } 2d = (2n+1) \lambda/2$$

◆ Measurement of Wavelength

When mirror M2 is moved by a distance x, let N fringes cross the field of view:

$$2x = N\lambda \rightarrow \lambda = 2x / N \leftarrow \text{Key Formula}$$

◆ Measurement of Wavelength Difference (λ_1 and λ_2)

If two close wavelengths are present, the fringes alternately appear distinct (maximum visibility) and indistinct (minimum visibility). If x is the path difference between two consecutive maximum visibility positions:

$$\Delta\lambda = \lambda^2 / (2x)$$

■ 7. Nature of Fringes

Condition	Type of Fringes	Shape
M1 ■ M2 (exactly parallel)	Fringes of equal thickness	Straight, parallel fringes
M1 slightly tilted w.r.t M2	Fringes of equal inclination	Circular fringes (rings)
White light used	Coloured fringes	Coloured straight fringes

■ 8. Advantages

- Extremely precise — can measure distances up to nanometre accuracy.
- Can measure both wavelength and wavelength difference.
- Can be used for both monochromatic and white light.

■ 9. Disadvantages

- Very sensitive to vibrations — needs vibration-free table.

- Requires highly skilled setup and alignment.
- Expensive and complex instrument.

■ 10. Applications

- Measurement of wavelength of monochromatic light: $\lambda = 2x/N$
- Measurement of difference between two closely spaced wavelengths.
- Measurement of refractive index of gases and transparent materials.
- Detection of gravitational waves (LIGO uses Michelson interferometer principle).
- Standardization of metre in terms of wavelength of light.
- Testing flatness of optical surfaces.

■ 11. Important Exam Points

- ★ $\lambda = 2x/N$ — most important formula for wavelength measurement.
- ★ Works on principle of Division of Amplitude.
- ★ G2 (compensating plate) is needed to equalize the optical path in glass.
- ★ Circular fringes → M1 and M2 exactly perpendicular; Straight fringes → slight tilt.

■ 12. Conclusion

The Michelson Interferometer is one of the most important and precise optical instruments ever built. It uses division of amplitude to create two coherent beams, and by analyzing their interference pattern, it enables highly accurate measurements of wavelength, wavelength difference, and refractive index. Its modern descendant, LIGO, detected gravitational waves using this very principle.

Q4. Nd:YAG Laser — Principle, Construction, Working, Energy Levels and Applications [13 Marks]

1. Definition

Nd:YAG Laser (Neodymium-doped Yttrium Aluminium Garnet Laser) is a solid-state laser that uses Nd^{3+} ions doped in YAG crystal as the active medium. It emits infrared light at a wavelength of 1064 nm and is one of the most widely used lasers in industry, medicine, and research.

2. Principle

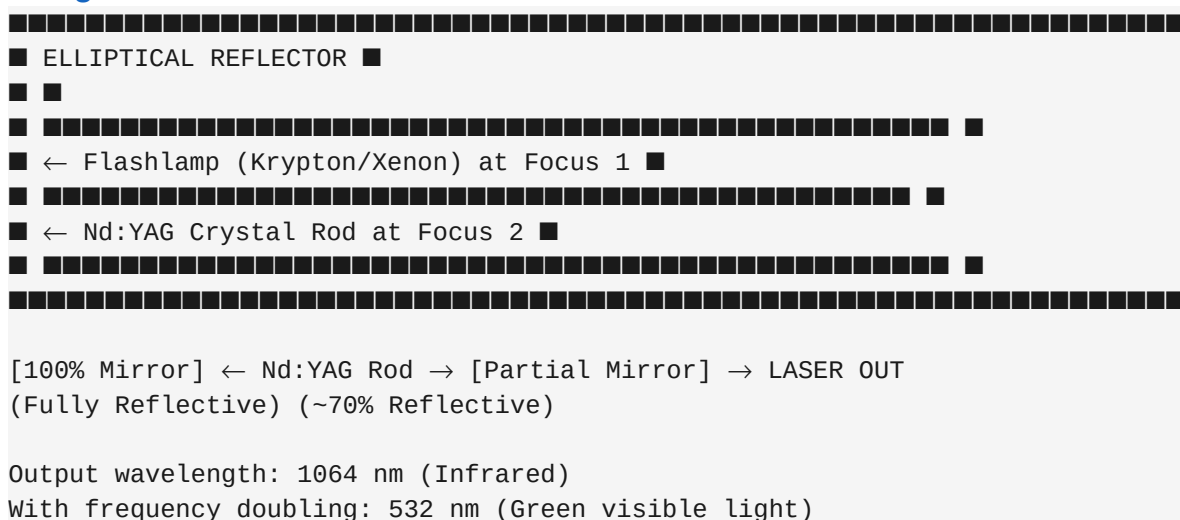
Nd:YAG Laser works on the principle of **Stimulated Emission of Radiation** (the same principle behind all lasers). The Nd^{3+} ions in the YAG crystal are excited to a higher energy level by optical pumping (using a flashlamp or diode laser). When stimulated by a photon, excited ions emit identical photons → laser action.

It is a **4-level laser** — meaning lasing occurs between two intermediate energy levels, not the ground state. This makes it much easier to achieve population inversion.

3. Construction

- **Active Medium:** Nd:YAG Crystal rod — YAG ($\text{Y}_3\text{Al}_2\text{O}_7$) doped with about 1% Nd^{3+} ions. Rod is typically 6–9 mm diameter and 5–20 cm long.
- **Pumping Source:** Krypton or xenon flashlamp (for pulsed operation) or diode laser (for continuous wave CW operation). Lamp is placed parallel to the crystal rod.
- **Elliptical Reflector:** Both the crystal rod and the lamp are placed at the two focal points of an elliptical reflector to maximize light collection.
- **Optical Resonator:** Two mirrors — Fully reflective mirror ($R=100\%$) at one end and Partially reflective mirror (output coupler, $\sim 50\text{--}80\%$) at other end.
- **Cooling System:** Water cooling to prevent overheating of the crystal.
- **Q-switch (optional):** For pulsed high-power output.

4. Diagram — Nd:YAG Laser



5. Energy Level Diagram — 4 Level System

Energy

Level 4 (E4) – Upper pump band (absorption)
 ↑ Fast non-radiative decay
 Pump ↓
 (flashlamp) Level 3 (E3) – Metastable (Upper laser level)
 Stimulated emission → LASER
 ↓ (1064 nm photon emitted)
 Level 2 (E2) – Lower laser level
 Fast non-radiative decay ↓
 Level 1 (E1) – Ground State

Population inversion between E3 and E2
 (4-level system → easy to achieve population inversion)

■ 6. Working

Step 1 — Pumping: Flashlamp emits intense white light. Nd^{3+} ions absorb light and jump from ground state (E1) to pump band (E4).

Step 2 — Fast Decay: Ions quickly fall from E4 to E3 (metastable state) by non-radiative transition (phonon emission). This is very fast ($\sim 10^{-10}$ s).

Step 3 — Population Inversion: Ions accumulate in E3 (long lifetime $\sim 230 \mu\text{s}$). Ions in E2 quickly drain to ground state. So $N(\text{E3}) \gg N(\text{E2}) \rightarrow$ Population Inversion achieved.

Step 4 — Stimulated Emission: A photon of 1064 nm triggers stimulated emission $\rightarrow \text{Nd}^{3+}$ ion drops from E3 to E2, emitting an identical photon.

Step 5 — Lasing: Photons bounce between mirrors $\rightarrow$ chain reaction $\rightarrow$ intense, coherent laser beam exits through partial mirror.

■ 7. Important Parameters

- Wavelength: **1064 nm** (infrared)
- With frequency doubling (SHG): **532 nm** (green), 355 nm (UV), 266 nm (deep UV)
- Mode: Pulsed or Continuous Wave (CW)
- Energy per pulse: Up to several Joules

■ 8. Advantages

- High power output possible.
- 4-level system — population inversion easy to achieve.
- Can operate in pulsed and CW modes.
- With frequency doubling, visible and UV light can be produced.
- YAG crystal is mechanically strong and thermally stable.

■ 9. Disadvantages

- Requires water cooling.
- Flashlamps need periodic replacement.
- Output is in infrared — not directly visible.

■ 10. Applications

- **Medical:** Eye surgery (retinal photocoagulation), kidney stone removal (lithotripsy), cancer treatment.
- **Industrial:** Laser cutting, welding, drilling of metals and ceramics.
- **Military:** Range finding, target designation.
- **Research:** Spectroscopy, laser cooling, non-linear optics.
- **Cosmetics:** Tattoo removal, skin resurfacing.

■ 11. Important Exam Points

- ★ **Nd:YAG is a 4-level SOLID STATE laser.**
- ★ **Active medium: Nd^{3+} ions in YAG crystal.**
- ★ **Output wavelength: 1064 nm (IR).**
- ★ **Pumping: Flashlamp or diode laser.**
- ★ **Population inversion is between E3 (metastable) and E2 (lower laser level).**

■ 12. Conclusion

Nd:YAG Laser is one of the most versatile and powerful solid-state lasers. Its 4-level energy scheme makes population inversion easy. With a wavelength of 1064 nm and frequency doubling options, it finds wide applications in industry, medicine, military, and research.


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| Discharge ■■■■ CO■ (00°1) – Asymmetric stretch mode (Upper laser level)
| Stimulated emission → 10.6 μm or 9.6 μm
| ↓
| ■■■■ CO■ (10°0) – Symmetric stretch mode (Lower laser level)
| Fast relaxation via He collisions
| ↓
| ■■■■ CO■ Ground State (00°0)
|
Lasing transition: 00°1 → 10°0 (10.6 μm) [most common]
or 00°1 → 02°0 (9.6 μm)

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■ 6. Working

Step 1: Electrical discharge through gas mixture excites N■ molecules to vibrational level ($v=1$).

Step 2: N■ ($v=1$) molecules collide with CO■ molecules and transfer energy (resonant energy transfer — both levels almost equal energy). CO■ is excited to asymmetric stretch mode 00°1 (upper laser level).

Step 3: Population inversion is established between 00°1 and 10°0 levels.

Step 4: Stimulated emission occurs — CO■ drops from 00°1 to 10°0, emitting photon of wavelength **10.6 μm**.

Step 5: Helium gas rapidly depopulates the lower level (10°0) by collisions, maintaining population inversion.

Step 6: Photons bounce in optical cavity → intense, coherent IR laser beam exits at 10.6 μm.

■ 7. Important Parameters

- Wavelength: **10.6 μm** (mid-infrared) — most common; also 9.6 μm
- Mode: Continuous Wave (CW) or Pulsed
- Efficiency: ~15–20% (very high for a laser!)
- Power: From watts to tens of kilowatts

■ 8. Advantages

- Highest power among gas lasers (kW range in CW).
- Very high efficiency (~15–20%).
- Can operate in both CW and pulsed modes.
- Low beam divergence.

■ 9. Disadvantages

- Output is in far infrared — invisible and absorbed by glass/air.
- Requires special IR-transparent optics (Ge, ZnSe).
- Cannot be transmitted through optical fibre easily.
- Requires large and bulky setup.

■ 10. Applications

- **Industrial:** Cutting, welding, engraving, and drilling of metals and non-metals.
- **Medical:** Surgery (skin resurfacing, scar removal), dental treatment.
- **Military:** Laser range finders, directed energy weapons.
- **Communications:** Atmospheric optical communications.
- **Research:** Molecular spectroscopy, isotope separation.

■ 11. Important Exam Points

- ★ CO₂ laser is a MOLECULAR gas laser (vibrational transitions).
- ★ Active medium: CO₂ + N₂ + He mixture.
- ★ Output: 10.6 μm (infrared).
- ★ N₂ transfers energy to CO₂ by resonant collision.
- ★ He helps remove population from lower laser level.
- ★ Highest efficiency gas laser (~15–20%).

■ 12. Conclusion

CO₂ Laser is the most powerful and efficient gas laser. It operates on vibrational transitions in CO₂ molecules and produces infrared radiation at 10.6 μm . Its high power makes it ideal for industrial cutting and welding, while its precision makes it useful in surgical procedures.

Q6. Comparison of Solid State, Gas and Semiconductor Laser Systems [13 Marks]

1. Introduction

Lasers are classified based on the state of matter of the active medium into: (1) Solid State Lasers, (2) Gas Lasers, and (3) Semiconductor (Diode) Lasers. Each type has distinct characteristics, advantages, and applications. Understanding their comparison is essential for selecting the right laser for a given application.

2. Solid State Laser — Overview

◆ Examples: Ruby Laser, Nd:YAG Laser, Nd:Glass Laser

- **Active Medium:** Solid crystal or glass doped with impurity ions (e.g., Cr^{3+} in Ruby; Nd^{3+} in YAG glass).
- **Pumping:** Optical pumping by flashlamp or laser diode.
- **Output:** High power, pulsed or CW. Wavelengths in visible to near-IR.
- **Energy Levels:** 3-level (Ruby) or 4-level (Nd:YAG) system.
- **Efficiency:** ~0.1–5%

3. Gas Laser — Overview

◆ Examples: He-Ne Laser (neutral atom), CO₂ Laser (molecular), Ar⁺ Laser (ion)

- **Active Medium:** Gas or mixture of gases at low pressure.
- **Pumping:** Electrical discharge (DC/RF), collisional pumping.
- **Output:** High coherence, narrow linewidth, mostly CW. Various wavelengths.
- **Efficiency:** ~0.01–20% (CO₂ is exceptionally high).

4. Semiconductor (Diode) Laser — Overview

◆ Examples: GaAs Laser, InGaAsP Laser, AlGaAs Laser

- **Active Medium:** p-n junction of semiconductor material.
- **Pumping:** Electrical (forward biased p-n junction — injection pumping).
- **Output:** Low to moderate power, pulsed or CW. Wavelength depends on bandgap.
- **Efficiency:** Very high — up to 70–80%!

5. Detailed Comparison Table

Parameter	Solid State Laser	Gas Laser	Semiconductor Laser
Active Medium	Solid crystal/glass + dopant	Gas or gas mixture	p-n junction of semiconductor
Example	Ruby, Nd:YAG	He-Ne, CO ₂ , Ar ⁺	GaAs, InGaAsP
Pumping Method	Optical (flashlamp/diode)	Electrical discharge	Electrical (current injection)
Wavelength Range	Visible to near-IR (e.g. 694 nm, 1064 nm)	UV to far-IR (e.g. 633 nm, 10.6 μm)	Near-IR to visible (e.g. 800–1550 nm)

Output Power	Very high (kW in pulsed)	High (CW kW for CO ₂)	Low to moderate (mW–W)
Efficiency	~0.1–5%	~0.01–20%	~30–80% (very high!)
Beam Quality	Very good	Excellent (very coherent)	Moderate (wide beam divergence)
Size	Medium to large	Large (long tube)	Very small (mm size)
Cost	High	Moderate to high	Very low
Cooling Required	Yes (water cooling)	Yes (water/forced air)	Minimal (small heat sink)
Operation Mode	Pulsed / CW	Mostly CW / pulsed	CW / pulsed
Applications	Surgery, material processing, range finding	Holography, spectroscopy, cutting	Optical fibre comms, CD/DVD, laser printers

■ 6. Energy Level Comparison

Laser Type	Energy Level System	Comment
Ruby Laser (Solid State)	3-level system	Harder to achieve population inversion; threshold is higher
Nd:YAG Laser (Solid State)	4-level system	Easy population inversion; lower threshold; more efficient
He-Ne Gas Laser	4-level system	CW, very coherent, low power
CO ₂ Gas Laser	Vibrational levels (4-level)	Highest power gas laser, IR output
Semiconductor Laser	Band-to-band (conduction to valence band)	Direct bandgap material needed (GaAs)

■ 7. Important Exam Points

- ★ Semiconductor laser has the **HIGHEST** efficiency (~70–80%).
- ★ Gas laser (He-Ne) has the **BEST** beam quality (most coherent).
- ★ Solid state laser gives the **HIGHEST POWER** in pulsed mode.
- ★ Semiconductor laser is the **SMALLEST** and **CHEAPEST** — used in CD/DVD players.
- ★ CO₂ laser has ~15–20% efficiency — highest among gas lasers.

■ 8. Conclusion

Each type of laser — solid state, gas, and semiconductor — has unique advantages suitable for specific applications. Solid state lasers provide high power, gas lasers provide excellent coherence, and semiconductor lasers provide miniaturization and high efficiency. Selecting the right laser depends on power requirement, wavelength, cost, and application.

■ 1. Definition

■ 2. Principle — Total Internal Reflection (TIR)

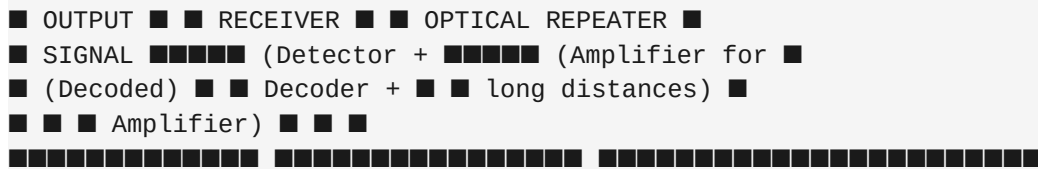
TIR occurs when light travels from a denser medium (high refractive index n_1 — core) to a less dense medium (lower refractive index n_2 — cladding) at an angle greater than or equal to the critical angle (θ_c).

When the angle of incidence at the core-cladding interface exceeds θ_c , the light is totally reflected back into the core and propagates along the fibre without loss.

■ 3. Structure of Optical Fibre

■ 4. Block Diagram — Optical Fibre Communication System

[illegible]



■ 5. Detailed Working — Each Block Explained

◆ Block 1: Input Signal (Information Source)

The original information to be transmitted — voice, video, computer data, etc. — is in electrical form (analog or digital).

◆ Block 2: Transmitter (Encoder + Light Source)

- **Encoder/Modulator:** Converts the electrical signal into a digital bit stream (modulation).
- **Light Source:** LED (for short distance, low bandwidth) or Laser Diode (for long distance, high bandwidth) converts electrical signal to light pulses.
- LED wavelengths: 850 nm, 1310 nm, 1550 nm.

◆ Block 3: Optical Fibre (Transmission Medium)

- Light pulses travel through the fibre by Total Internal Reflection.
- Signal loss occurs due to: Absorption, Scattering (Rayleigh), and Bending losses.
- Minimum loss window: 1550 nm (0.2 dB/km).

◆ Block 4: Optical Repeater/Amplifier (for long distances)

For very long distances (>100 km), EDFA (Erbium Doped Fibre Amplifier) boosts the optical signal without converting to electrical form.

◆ Block 5: Receiver (Photodetector + Decoder)

- **Photodetector:** PIN Photodiode or Avalanche Photodiode (APD) converts received light pulses back to electrical signal.
- **Amplifier:** Amplifies the weak electrical signal.
- **Decoder:** Recovers original information from the digital signal.

■ 6. Types of Optical Fibres

Type	Core Dia.	Feature	Use
Single Mode (SM)	~8–10 μm	One mode, lowest loss, highest bandwidth	Long-distance telecom
Multi Mode Step-Index	~50–200 μm	Many modes, modal dispersion	Short distance, LAN
Multi Mode Graded-Index	~50 μm	Reduced dispersion, good bandwidth	Medium distance

■ 7. Numerical Aperture (NA)

NA measures the light-gathering ability of the fibre:

$$NA = \sin(\theta_{\max}) = \sqrt{(n_{\text{core}}^2 - n_{\text{clad}}^2)}$$

Where $\theta_{\max}$ = acceptance angle; n_{core} = core index; n_{cladding} = cladding index.

■ 8. Advantages

- Very low signal loss (0.2 dB/km at 1550 nm).
- Extremely large bandwidth (Tbps possible).
- No electromagnetic interference (EMI).
- Lightweight and thin.
- Secure — cannot be tapped easily.
- No electrical sparks — safe in hazardous environments.

■ 9. Disadvantages

- Brittle — cannot be bent sharply.
- Splicing and connectors are expensive and skilled work.
- Cannot carry electrical power.

■ 10. Applications

- Long-distance telecommunications (internet backbone, telephone).
- Local Area Networks (LAN) and data centres.
- Medical endoscopy (viewing internal body parts).
- Sensing applications (temperature, pressure sensors).
- Military and aerospace secure communications.

■ 11. Important Exam Points

- ★ **Principle: Total Internal Reflection (TIR).**
- ★ **$NA = \sqrt{(n_1^2 - n_2^2)}$ — learn this formula.**
- ★ **Minimum loss: 1550 nm wavelength.**
- ★ **Single mode fibre: 8–10 μm core — used for long distance.**
- ★ **EDFA = Erbium Doped Fibre Amplifier — used for optical amplification.**

■ 12. Conclusion

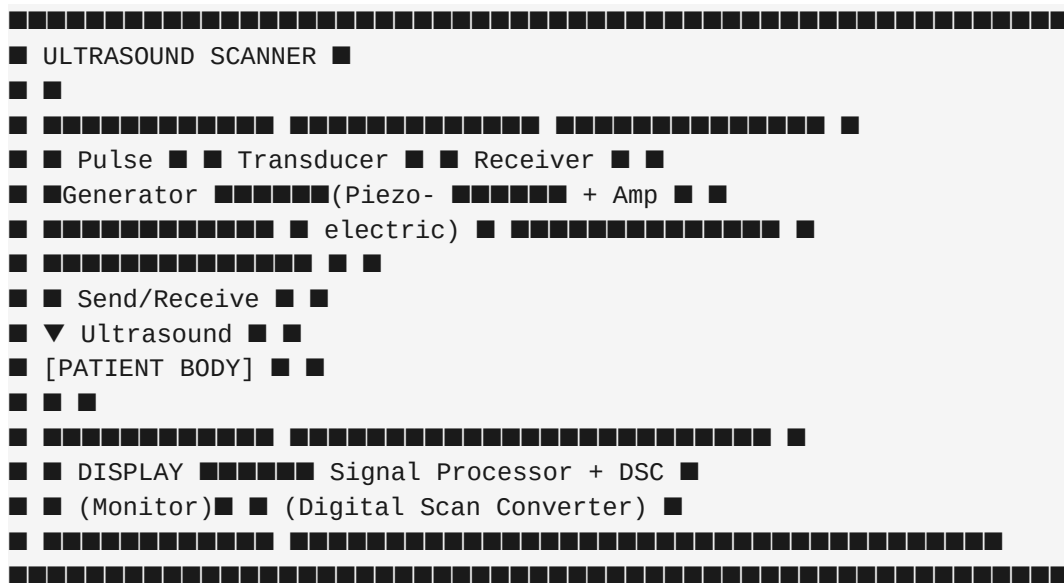
Optical Fibre Communication is the backbone of modern internet and telecommunications. Using Total Internal Reflection, it transmits information as light pulses with minimal loss, enormous bandwidth, and complete immunity to electromagnetic interference. It has revolutionized global communications.

■ 1. Definition

■ 2. Principle

- A transducer emits short pulses of ultrasound waves (frequency 1–20 MHz) into the body.
- These waves travel through tissues and are partially **reflected (echoed)** at interfaces between different tissue types (bone, muscle, fluid, fat).
- The transducer detects the returning echoes.
- The **time delay** and **amplitude** of echoes give the distance and nature of the reflecting surface.
- A computer processes these echoes and constructs a 2D image.

■ 3. Block Diagram — Ultrasound Imaging System



■ 4. Types of Ultrasound Imaging Modes

◆ A-Mode (Amplitude Mode)

- ```
Display:
Amplitude
| | | | |
```

|\_|\_|\_|\_|\_|\_|\_|\_|→ Time/Depth  
Transducer Body surfaces / interfaces

### ◆ B-Mode (Brightness Mode) — Most Common

- Each echo is displayed as a **bright dot**, with brightness proportional to echo amplitude.
- As the transducer is scanned, multiple lines build up a **2D cross-sectional image**.
- Modern scanners capture many frames per second → **Real-time imaging**.
- Used in: Obstetrics (baby monitoring), abdominal scans, cardiac imaging.

Display:

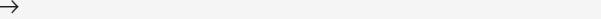
[illegible]

### ◆ M-Mode (Motion Mode)

- Records the **motion of structures over time**.
- A single line of B-mode echoes is scrolled over time on the y-axis.
- Used in: **Echocardiography** — studying heart valve motion and wall thickness.

Display:

Time →



Shows rhythmic motion of heart walls/valves

## ◆ Doppler Mode

- Uses **Doppler effect** — frequency shift of echoes from moving objects (e.g., blood cells).
- Moving blood cells cause a frequency shift:  $\Delta f = 2fv \cos \theta / c$
- Used to measure **blood flow velocity and direction**.
- Colour Doppler: flow towards transducer shown in RED; away in BLUE.
- Used in: Cardiac blood flow assessment, detection of blocked arteries.

## ■ 5. Transducer — Key Component

- Made of **piezoelectric crystal** (e.g., PZT — Lead Zirconate Titanate).
- **Transmitter:** Applied alternating voltage → crystal vibrates → emits ultrasound.
- **Receiver:** Returning echo → vibrates crystal → generates electrical signal.
- Same transducer works as both transmitter and receiver (pulse-echo mode).

## ■ 6. Comparison of Imaging Modes

| Mode    | Display                       | Application              |
|---------|-------------------------------|--------------------------|
| A-Mode  | Amplitude vs Depth (1D spike) | Eye measurements         |
| B-Mode  | 2D brightness image           | Abdominal, fetal imaging |
| M-Mode  | Motion vs Time                | Echocardiography         |
| Doppler | Frequency shift / Colour map  | Blood flow measurement   |

## ■ 7. Advantages

- Non-ionizing (no X-ray radiation) — completely safe.
- Real-time imaging possible.
- Portable and relatively inexpensive.
- Can image soft tissues (unlike X-ray).

## ■ 8. Disadvantages

- Cannot penetrate bone or gas (air) — poor for lung/bone imaging.
- Operator-dependent — skill needed.
- Limited depth penetration for high frequencies.

## ■ 9. Applications

- Obstetrics — fetal development monitoring, detecting anomalies.
- Cardiology — echocardiography, blood flow measurement.
- Abdominal imaging — liver, kidney, gallbladder.
- Emergency medicine — FAST (Focused Assessment with Sonography in Trauma).
- Non-destructive testing (NDT) of materials in industry.

## ■ 10. Important Exam Points

- ★ **A-Mode: 1D amplitude display; B-Mode: 2D image; M-Mode: motion; Doppler: blood flow.**
- ★ **Principle: Pulse-Echo method.  $d = vt/2$ .**
- ★ **Transducer: Piezoelectric crystal (PZT).**
- ★ **Frequency used: 1–20 MHz (medical applications).**

## ■ 11. Conclusion

*Ultrasound imaging is a safe, real-time, and versatile medical imaging technique. Using the pulse-echo principle, it generates A-mode, B-mode, M-mode, and Doppler images that help diagnose a wide range of medical conditions. Its non-ionizing nature makes it the preferred imaging method for pregnant women and routine scans.*

## Q9. Derivation of Time-Dependent Schrödinger Wave Equation [13 Marks]

### 1. Introduction

The Schrödinger Wave Equation is the fundamental equation of Quantum Mechanics, analogous to Newton's second law in classical mechanics. It was proposed by Erwin Schrödinger in 1926. It describes how the quantum state (wave function  $\psi$ ) of a physical system changes with time and space.

### 2. Basic Concepts

- In quantum mechanics, particles behave as waves (de Broglie hypothesis).
- A particle is described by its **wave function**  $\psi(x, y, z, t)$ .
- $|\psi|^2$  = probability density of finding the particle at that point.
- The Schrödinger equation governs the evolution of  $\psi$  with time.

### 3. Derivation of Time-Dependent Schrödinger Equation

#### ◆ Step 1: Start with the de Broglie Wave

A free particle moving in the x-direction can be represented as a plane wave:

$$\psi(x,t) = A \cdot \exp[i(kx - \omega t)]$$

Where:  $k = 2\pi/\lambda$  (wave number),  $\omega = 2\pi\nu$  (angular frequency)

#### ◆ Step 2: Total Energy Relation

From Planck's quantum theory, total energy  $E$  of a particle:

$$E = h\nu = \hbar\omega \text{ (where } \hbar = h/2\pi)$$

From de Broglie: momentum  $p = h/\lambda = \hbar k$

Total energy = Kinetic Energy + Potential Energy:

$$E = p^2/(2m) + V(x,t)$$

#### ◆ Step 3: Differentiate $\psi$ with respect to time

$$\partial\psi/\partial t = -i\omega \cdot A \cdot \exp[i(kx - \omega t)] = -i\omega \cdot \psi$$

Since  $E = \hbar\omega$ , we have  $\omega = E/\hbar$ :

$$\partial\psi/\partial t = -i(E/\hbar) \cdot \psi$$

Rearranging:

$$E \cdot \psi = i\hbar \cdot \partial\psi/\partial t \dots (1)$$

#### ◆ Step 4: Differentiate $\psi$ with respect to x (twice)

$$\partial\psi/\partial x = ik \cdot \psi$$

$$\partial^2\psi/\partial x^2 = (ik)^2 \cdot \psi = -k^2 \cdot \psi$$

Since  $p = \hbar k$ , we have  $k^2 = p^2/\hbar^2$ :

$$\partial^2\psi/\partial x^2 = -(p^2/\hbar^2) \cdot \psi$$

Therefore:

$$p^2 \cdot \psi = -\hbar^2 \cdot \partial^2\psi/\partial x^2 \dots (2)$$

#### ◆ Step 5: Substitute into Energy Equation

Multiply total energy equation  $E = p^2/2m + V$  by  $\psi$ :

$$E \cdot \psi = (p^2/2m) \cdot \psi + V \cdot \psi$$

Substituting from (1) and (2):

$$i\hbar \partial\psi/\partial t = -(\hbar^2/2m) \partial^2\psi/\partial x^2 + V \cdot \psi$$

#### ■ 4. Final Form — Time-Dependent Schrödinger Equation (1D)

$i\hbar \partial\psi/\partial t = -(\hbar^2/2m) \partial^2\psi/\partial x^2 + V(x,t) \cdot \psi$  This is the Time-Dependent Schrödinger Wave Equation in ONE dimension.

##### ◆ For 3D (General Form):

$$i\hbar \partial\psi/\partial t = -(\hbar^2/2m) \nabla^2\psi + V(\mathbf{r},t) \cdot \psi$$

Where  $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$  is the Laplacian operator.

This can be written compactly using the Hamiltonian operator  $\hat{H}$ :

$$i\hbar \partial\psi/\partial t = \hat{H} \cdot \psi$$

Where  $\hat{H} = -(\hbar^2/2m)\nabla^2 + V$  is the Hamiltonian operator.

#### ■ 5. Symbol Summary

| Symbol       | Meaning                              | Unit     |
|--------------|--------------------------------------|----------|
| $\psi$ (psi) | Wave function                        | —        |
| $ \psi ^2$   | Probability density                  | $m^{-3}$ |
| $\hbar$      | Reduced Planck constant ( $h/2\pi$ ) | J·s      |
| $m$          | Mass of particle                     | kg       |
| $V(x,t)$     | Potential energy                     | J        |
| $E$          | Total energy of particle             | J or eV  |
| $i$          | Imaginary unit ( $\sqrt{-1}$ )       | —        |

#### ■ 6. Physical Interpretation

- $\psi(x,t)$  is the probability amplitude of finding the particle at position  $x$  at time  $t$ .
- $|\psi(x,t)|^2$  = probability density — tells us where the particle is most likely to be found.
- The Schrödinger equation is a LINEAR, FIRST-ORDER equation in time.
- It is NOT derived from first principles — it is a postulate of quantum mechanics.

#### ■ 7. Special Case — Time-Independent Schrödinger Equation

When  $V$  is independent of time, we can separate  $\psi(x,t) = \phi(x) \cdot T(t)$ :

$$T(t) = \exp(-iEt/\hbar)$$

And  $\phi(x)$  satisfies the Time-Independent Schrödinger Equation:

$$-(\hbar^2/2m) d^2\phi/dx^2 + V(x) \cdot \phi = E \cdot \phi$$

#### ■ 8. Important Exam Points

★ Full equation:  $i\hbar \partial\psi/\partial t = -(\hbar^2/2m) \partial^2\psi/\partial x^2 + V\psi$

- ★ It is a POSTULATE — not derived from other physical laws.
- ★  $|\psi|^2$  gives probability density, NOT  $\psi$  itself.
- ★ Wave function must be: single-valued, continuous, finite, and normalizable.
- ★ For stationary states:  $\psi(x,t) = \phi(x) \cdot e^{(-iEt/\hbar)}$ .

## ■ 9. Conclusion

*The Time-Dependent Schrödinger Wave Equation  $i\hbar \partial\psi/\partial t = -(\hbar^2/2m)\nabla^2\psi + V\psi$  is the cornerstone of quantum mechanics. Starting from the de Broglie wave and classical energy relation, we derived this equation that governs all quantum mechanical systems. The wave function  $\psi$  encodes all information about the quantum state of a particle.*

## ■ 1. Introduction

## ■ 2. Problem Setup — The Infinite Potential Well

- Inside the box ( $0 < x < L$ ):  $V = 0$  (particle moves freely)
- Outside the box ( $x \leq 0$  and  $x \geq L$ ):  $V = \infty$  (particle cannot escape)

Particle is confined in  $0 < x < L$   
 $\psi = 0$  outside the box (particle cannot be there)

◆ Step 1: Apply Time-Independent Schrödinger Equation inside the box ( $V=0$ )

$$\text{Let } k^2 = 2mE/\hbar^2 \rightarrow d^2\psi/dx^2 + k^2\psi = 0$$

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

Since  $V = \infty$  outside, the wave function must be zero at the walls:

**At  $x = 0$ :**  $\psi(0) = A \cdot \sin(0) + B \cdot \cos(0) = B = 0 \rightarrow B = 0$

So:  $\psi(x) = A \sin(kx)$

**At  $x = L$ :**  $\psi(L) = A \sin(kL) = 0$

Since  $A \neq 0$  (trivial solution), we need  $\sin(kL) = 0$ :

$$kL = n\pi \rightarrow k = n\pi/L \quad (n = 1, 2, 3, \dots)$$

### ◆ Step 4: Energy Quantization

Substituting  $k = n\pi/L$  into  $k^2 = 2mE/\hbar^2$ :

$$(n\pi/L)^2 = 2mE_n/\hbar^2$$

$$E_n = n^2\pi^2\hbar^2 / (2mL^2) = n^2h^2 / (8mL^2) \leftarrow \text{ENERGY IS QUANTIZED!}$$

This is the **Energy Eigenvalue**. Here:

- $n = 1, 2, 3, \dots$  (quantum number)
- $n = 1$ : Ground state energy  $E_1 = h^2/(8mL^2) = E_1$  (minimum energy)
- $n = 2$ :  $E_2 = 4E_1$ ;  $n = 3$ :  $E_3 = 9E_1$ ; etc.
- Energy is proportional to  $n^2$  — it cannot be zero!

#### ◆ Step 5: Wave Functions (Normalized)

Applying normalization condition  $\int |\psi|^2 dx = 1$  from 0 to L:

$$\psi_n(x) = \sqrt{(2/L)} \cdot \sin(n\pi x/L) \leftarrow \text{Normalized wave function}$$

### ■ 5. Diagram — Energy Levels and Wave Functions

Energy Wave functions  $\psi_n(x)$  Probability  $|\psi_n|^2$

$E_4 = 16E_1$   $\psi_4$ :   $n=4$  (4 loops)

$E_3 = 9E_1$   $\psi_3$ :   $n=3$  (3 loops)

$E_2 = 4E_1$   $\psi_2$ :   $n=2$  (2 loops)

$E_1 = 1E_1$   $\psi_1$ :   $n=1$  (1 loop – Ground State)

$x=0$   $x=L$

Key:  $E_1 = h^2/(8mL^2)$  – Zero-point energy ( $n=1$ , minimum energy)

Energy spacing increases as  $n$  increases:  $E_n = n^2E_1$

### ■ 6. Proof that Energy is Quantized

From the solution, we see that energy  $E_n = n^2h^2/(8mL^2)$  can ONLY take specific discrete values corresponding to  $n = 1, 2, 3, \dots$

It cannot take any intermediate value — this is the **quantization of energy**.

- $n=1$ :  $E_1 = h^2/8mL^2$  (lowest energy — ground state)
- $n=2$ :  $E_2 = 4h^2/8mL^2 = 4E_1$
- $n=3$ :  $E_3 = 9E_1$
- Energy is NOT continuous — it is QUANTIZED in units of  $E_1$ .

This is a purely quantum mechanical result — there is NO classical equivalent. Even the ground state ( $n=1$ ) has non-zero energy  $E_1$  called the **Zero-Point Energy**.

### ■ 7. Zero-Point Energy

In classical mechanics, a confined particle can have ZERO kinetic energy (at rest). In quantum mechanics, the minimum energy is:

$$E_1 \text{ (Zero-Point Energy)} = h^2 / (8mL^2) \neq 0$$

This is a direct consequence of Heisenberg's Uncertainty Principle — a perfectly still particle would have zero momentum uncertainty, violating the principle.

## ■ 8. Key Formulae Summary

| Quantity            | Formula                                                 |
|---------------------|---------------------------------------------------------|
| Allowed energies    | $E_n = n^2 h^2 / (8mL^2) = n^2 \pi^2 \hbar^2 / (2mL^2)$ |
| Wave functions      | $\psi_n = \sqrt{2/L} \cdot \sin(n\pi x/L)$              |
| Zero-point energy   | $E_1 = h^2 / (8mL^2)$                                   |
| Probability density | $ \psi_n ^2 = (2/L) \sin^2(n\pi x/L)$                   |

## ■ 9. Important Exam Points

- ★ **FINAL FORMULA:**  $E_n = n^2 h^2 / (8mL^2)$
- ★ Energy is **QUANTIZED** — only discrete values allowed.
- ★ Zero-point energy  $E_1 = h^2 / (8mL^2) \neq 0$  (quantum mechanical result).
- ★ Wave function:  $\psi_n = \sqrt{2/L} \sin(n\pi x/L)$
- ★  $n = 1$  is ground state;  $n$  cannot be 0 (that would give  $\psi = 0$  everywhere).
- ★ Energy levels:  $E_1 : E_2 : E_3 = 1 : 4 : 9$  (ratio of  $n^2$ ).

## ■ 10. Conclusion

*The particle in an infinite potential well is a foundational quantum mechanics problem. By applying the Schrödinger equation and boundary conditions, we proved rigorously that energy is quantized:  $E_n = n^2 h^2 / (8mL^2)$ . The existence of zero-point energy and the discrete energy spectrum are purely quantum phenomena with no classical counterpart, demonstrating the fundamental difference between classical and quantum mechanics.*

### Q11. Transmission Electron Microscope (TEM) — Construction and Working [13 Marks]

## ■ 1. Definition

A Transmission Electron Microscope (TEM) is a type of electron microscope that uses a beam of electrons transmitted through a very thin specimen to form a highly magnified image. It can achieve resolutions up to 0.05 nm (50 pm), allowing direct imaging of atoms, crystal structures, and cellular organelles — far beyond what light microscopes can achieve.

## ■ 2. Principle

TEM works on the principle of **Wave nature of electrons** (de Broglie's hypothesis).

Electrons, when accelerated through high voltage (60–300 kV), have a very short de Broglie wavelength (~0.004 nm at 100 kV), much smaller than the wavelength of visible light (~500 nm). This extremely small wavelength allows extremely high resolution.

de Broglie wavelength:  $\lambda = h/p = h/\sqrt{2meV}$

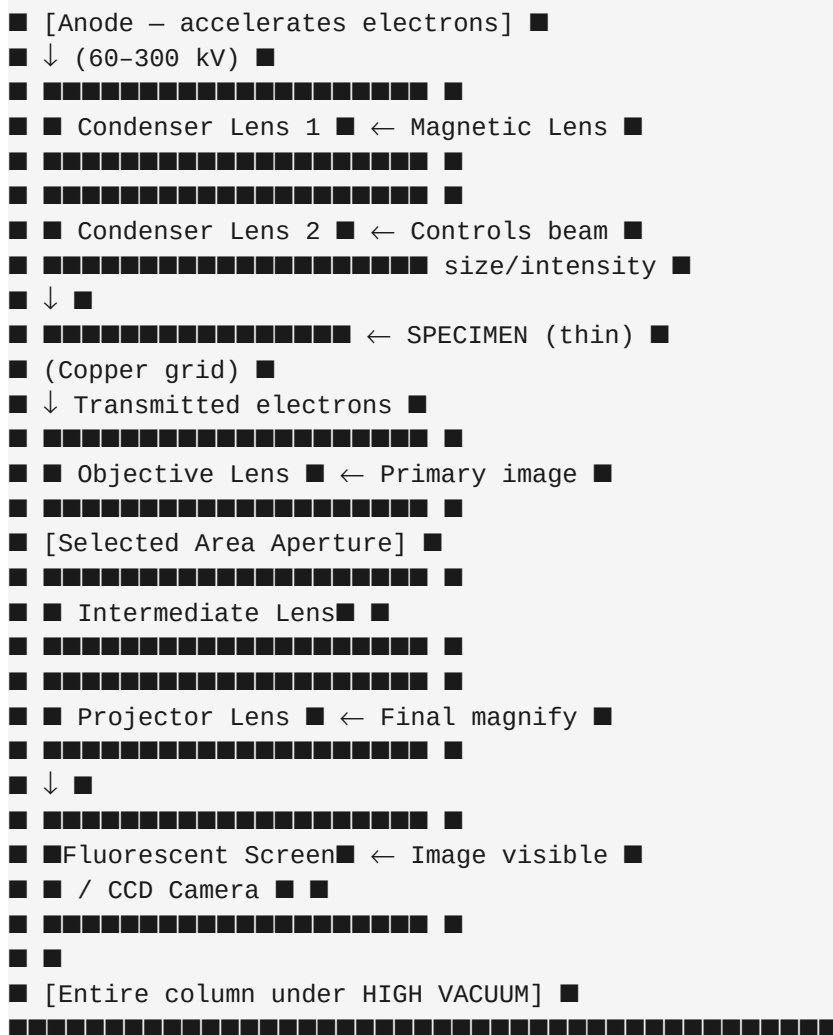
Electron lenses (magnetic lenses) focus the electron beam just like glass lenses focus light.

### ■ 3. Construction — Main Components

- **Electron Gun:** Produces a beam of electrons by thermionic emission from a heated tungsten filament (cathode). Anode accelerates electrons to 60–300 kV.
- **Condenser Lens System (2 lenses):** First condenser lens reduces the beam size; Second condenser lens controls the beam intensity and spot size on specimen.
- **Specimen Stage:** Ultra-thin specimen (< 100 nm thick) mounted on a copper grid, inserted into the electron beam path.
- **Objective Lens:** The most critical lens — forms the primary magnified image of the specimen. Determines the resolution.
- **Selected Area Aperture:** Selects specific region of specimen for imaging or diffraction.
- **Intermediate Lens:** Further magnifies the image from objective lens.
- **Projector Lens:** Final magnification — projects the electron image onto the viewing screen or camera.
- **Fluorescent Screen / CCD Camera:** Electrons hit fluorescent screen → visible image appears. Modern TEMs use CCD cameras for digital imaging.
- **Vacuum System:** Entire column is maintained under high vacuum ( $\sim 10^{-6}$  Pa) to prevent electron scattering by air molecules.

#### ■ 4. Diagram — TEM

[illegible]



## ■ 5. Working — Step by Step

**Step 1 — Electron Generation:** The tungsten filament (cathode) is heated to  $\sim 2700$  K. Electrons are emitted by thermionic emission.

**Step 2 — Acceleration:** The Wehnelt cylinder concentrates the electron beam. The anode accelerates electrons through 60–300 kV potential difference, giving them high energy and short wavelength.

**Step 3 — Condenser Lenses:** Two magnetic condenser lenses focus the electron beam into a fine, intense, parallel beam and illuminate the specimen.

**Step 4 — Interaction with Specimen:** The electron beam passes THROUGH the ultra-thin specimen. Denser regions of the specimen scatter more electrons (appear dark). Less dense regions allow more electrons through (appear bright). This creates contrast in the image.

**Step 5 — Objective Lens:** Collects transmitted electrons and forms the first magnified image (up to 100× magnification at this stage).

**Step 6 — Intermediate and Projector Lenses:** Further magnify the image. Total magnification can reach 1,000,000× (1 million times!).

**Step 7 — Image Formation:** Final electron image strikes the fluorescent screen → glows green/yellow, creating a visible image. Modern TEM uses CCD cameras for digital recording.

## ■ 6. Image Contrast in TEM

| Type of Contrast        | Cause                                                    | Used For                               |
|-------------------------|----------------------------------------------------------|----------------------------------------|
| Mass-Thickness Contrast | Heavier atoms/thicker regions scatter more electrons     | Biological samples, polymers           |
| Diffraction Contrast    | Crystal planes diffract electrons at specific angles     | Crystal defects, dislocations          |
| Phase Contrast          | Phase difference between scattered and unscattered beams | High-resolution atomic imaging (HRTEM) |

## ■ 7. Resolution of TEM

The resolution is determined by the de Broglie wavelength of electrons:

$$\lambda = h / \sqrt{2meV}$$

- At 100 kV:  $\lambda \approx 0.004 \text{ nm}$  → resolution  $\sim 0.1\text{--}0.2 \text{ nm}$
- At 300 kV: resolution better than  $0.05 \text{ nm}$  (can image individual atoms!)
- Comparison: Light microscope resolution  $\sim 200 \text{ nm}$

## ■ 8. Advantages

- Extremely high resolution — can image individual atoms and crystal planes.
- Very high magnification — up to  $1,000,000\times$ .
- Can perform electron diffraction to study crystal structure.
- Provides both image and diffraction pattern.

## ■ 9. Disadvantages

- Sample must be extremely thin ( $< 100 \text{ nm}$ ) — difficult preparation.
- Sample may be damaged by intense electron beam.
- Requires high vacuum — samples must be dry (no wet/living samples).
- Very expensive equipment.
- Requires specialized skill to operate and interpret images.
- 2D projection only — not directly 3D (though tomography is possible).

## ■ 10. Applications

- **Materials Science:** Crystal structure, grain boundaries, dislocations in metals.
- **Biology:** Virus structure, cell organelles (mitochondria, ribosomes).
- **Nanotechnology:** Nanoparticles, carbon nanotubes, quantum dots imaging.
- **Semiconductor:** IC device characterization, thin film analysis.
- **Forensic Science:** Identifying particles and fibres.

## ■ 11. Comparison — TEM vs Light Microscope

| Parameter      | TEM                                | Light Microscope                  |
|----------------|------------------------------------|-----------------------------------|
| Radiation used | Electron beam                      | Visible light                     |
| Wavelength     | $\sim 0.004 \text{ nm}$ (100 kV)   | $\sim 400\text{--}700 \text{ nm}$ |
| Resolution     | $\sim 0.05\text{--}0.2 \text{ nm}$ | $\sim 200 \text{ nm}$             |

|               |                          |                      |
|---------------|--------------------------|----------------------|
| Magnification | Up to 1,000,000×         | Up to ~2000×         |
| Lenses        | Magnetic lenses          | Glass lenses         |
| Vacuum        | Required ( $10^{-6}$ Pa) | Not required         |
| Sample        | Must be thin (<100 nm)   | Can be thick, living |

## ■ 12. Important Exam Points

- ★ TEM uses ELECTRONS, not light — very short wavelength → very high resolution.
- ★ Magnetic lenses (not glass) are used to focus electrons.
- ★ HIGH VACUUM is essential — air would scatter electrons.
- ★ Sample must be ultra-thin (< 100 nm).
- ★ Resolution ~ 0.05–0.2 nm; Magnification up to  $10^6\times$ .
- ★ Principle: de Broglie wave nature of electrons.

## ■ 13. Conclusion

*The Transmission Electron Microscope is the most powerful imaging tool available to scientists. Exploiting the extremely short de Broglie wavelength of high-energy electrons, it achieves sub-nanometre resolution — making it possible to directly image atoms and crystal structures. It is indispensable in materials science, biology, nanotechnology, and semiconductor research.*